Reference 07: Solar PV Lifecycle CO2 Emissions

1. What is this? (Metric & Checked Value)

- **Metric:** Lifecycle Greenhouse Gas Emissions of Solar Photovoltaics (Crystalline Silicon)
- **Checked Value:** $46\ \text{g}\ \text{CO}_2\text{e}/\text{kWh}$ (Median, with interquartile range of $35–65\ \text{g}\ \text{CO}_2\text{e}/\text{kWh}$)
- **Lifecycle Scope:** Raw silicon mining, wafer/cell fabrication, aluminum frame extrusion, transport, installation, and end-of-life recycling.

2. The Proof (Screenshot & Harmonized Findings)

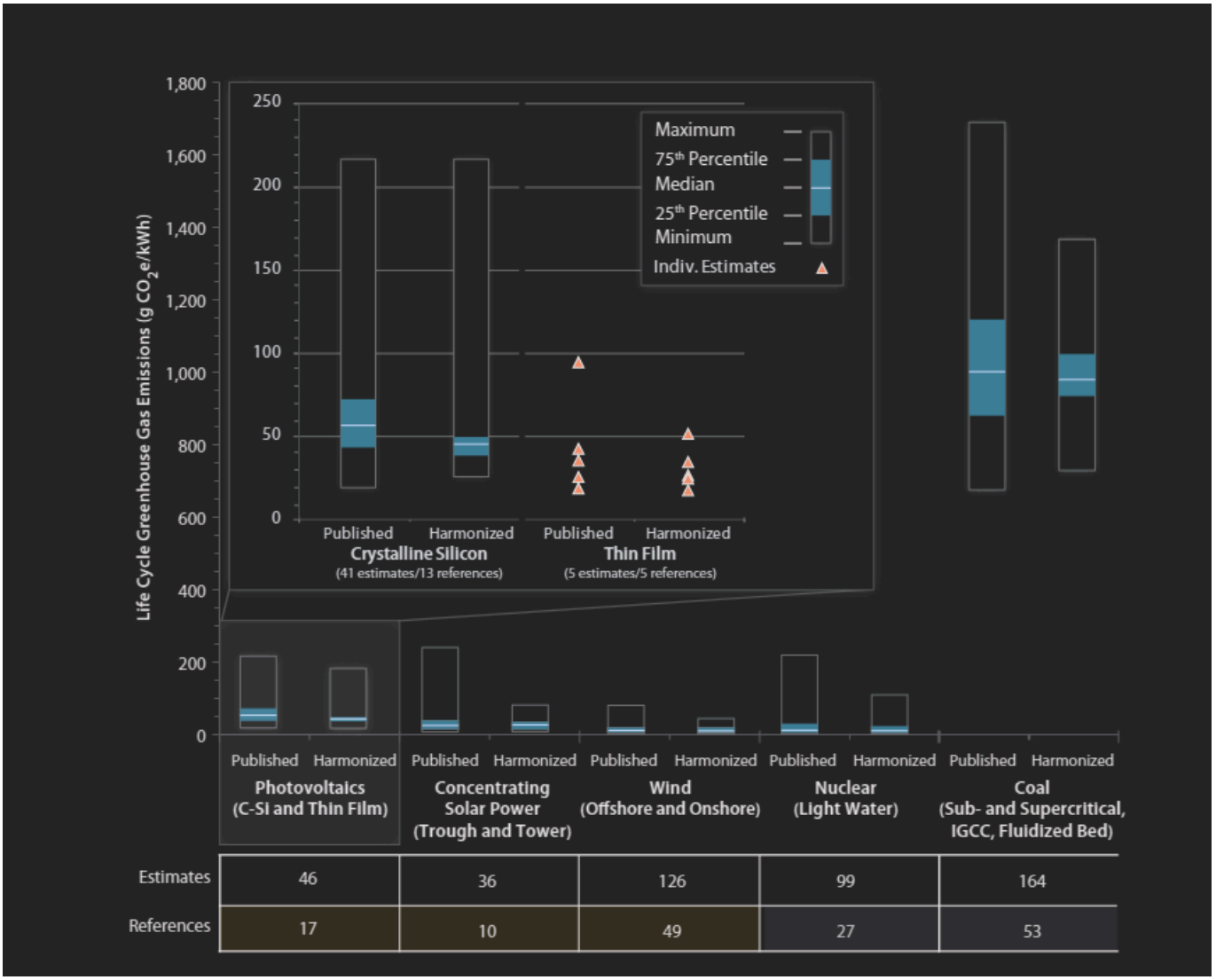


Figure 7: NREL Harmonized Life Cycle Greenhouse Gas Emissions from Utility and Distributed Solar Photovoltaics.

> **NREL Harmonized LCA Finding:** Across hundreds of published peer-reviewed lifecycle studies, crystalline silicon PV produces a median life cycle footprint of $43–48\ \text{g}\ \text{CO}_2\text{e}/\text{kWh}$ (standard reference value: $46\ \text{g}\ \text{CO}_2\text{e}/\text{kWh}$).

3. How was it calculated?

- **Clinic Annual Generation:** $5,840\ \text{kWh/year}$
- **Solar Lifecycle Footprint:**

$$\text{Solar } CO_2 = 5,840\ \text{kWh/yr} \times 0.046\ \text{kg } CO_2\text{/kWh} = \mathbf{0.269}\ \text{tonnes } \mathbf{CO_2}\text{/year} \approx \mathbf{0.27}\ \text{t/yr}$$

- **Over 20-Year Project Life:**

$$20\ \text{yrs} \times 0.269\ \text{t/yr} = \mathbf{5.37}\ \text{tonnes } \mathbf{CO_2}\ \text{total}$$

- **Comparison to Generative AI Guess:**
  - AI guessed $\approx 45–50\ \text{g } CO_2\text{/kWh}$ ($0.26–0.29\ \text{t/yr}$).
  - Confirms the AI's estimate is accurate and fully aligns with NREL's harmonized standard.

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4. Source Link

- **NREL Harmonized Life Cycle Assessment:** NREL Lifecycle Greenhouse Gas Emissions from Solar PV
- **Technical Report:** Life Cycle Greenhouse Gas Emissions from Utility-Scale and Distributed Photovoltaics (PDF)